QUEENSLAND UNIVERSITY OF TECHNOLOGY
Faculty of Engineering

EGH490-1 — Research Project 1
Project Progress Report

Design and Development of a Bidirectional DC–DC Power Shuffler for Partial Shading Mitigation in Photovoltaic Systems

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Contents

1. Project Summary 3
2. Interim Deliverables 4
2.1 Circuit Simulations 4
2.2 Review and Identified Challenges 10
2.3 Limitations and Assumptions 10
3. Risk Management 11
Project Risks 11
Experimental Risks 11
4. Consideration of Ethics 12
Demonstrate Integrity 12
Practise Competently 12
Exercise Leadership 13
Promote Sustainability 13
Intellectual Property Considerations 13
5. Implementation of Sustainable Development Principles 13
System-Level Sustainability through Improved Energy Yield 14
Component Selection for Longevity and Repairability 14
Modular Hardware Strategy 14
Standardised Interfaces and Long-Term Compatibility 14
6. Resources, Codes, and Standards 14
Software 14
Hardware and Facilities 15
Codes and Standards 15
7. Stakeholder Engagement 15
8. Conclusion 16
References 16
Appendix A — Updated Timeline Status 17

1. Project Summary

Within the last decade, photovoltaic (PV) cells have become a dominant method of power generation in the renewable energy sector. Despite improvements in PV cell efficiency, the electrical layout within a module has remained roughly unchanged: several series PV cell strings connected in parallel. This layout is susceptible to the effects of partial shading; if a cell in a string is shaded, the output current of the entire string is limited to that of the shaded cell. In severe shading cases, the cell becomes reverse-biased and acts as a resistor, leading to the formation of hotspots which can damage neighbouring cells. To rectify this problem, bypass diodes are typically connected antiparallel between every 6–8 cells in a string, restoring current flow if one or a few cells are unevenly shaded. Whilst this mitigation strategy is simple and cost-effective, it results in total loss of energy generated by cells or modules between two bypass diodes and introduces multiple maximum power points (MPPs), leading to ineffective energy extraction by conventional maximum power point trackers (MPPTs). This problem also extends to PV strings with curvature, as they will receive non-uniform illuminance across individual module, leading to easier activation of the bypass diodes.

To overcome the limitations of the bypass diode, a new approach in the form of 'shuffling converters' is proposed. Shuffling converters allow for current redistribution in a string of PV modules, where current from unshaded modules is transferred to shaded modules, restoring current flow and preventing activation of the bypass diode. This corrective action allows for greater energy extraction efficiency from a partially shaded PV string and prevents the formation of multiple MPPs, allowing for more effective tracking by MPPT controllers. This type of converter falls under the category of Serial connected Partial Power Converter (S-PPC).

This research project aims to investigate the viability of implementing a shuffling DC–DC converter in a PV string in combination with a diode–capacitor pair used for MPPT of the array. Whilst research into this topic has been conducted, there is an overall lack of discussion in the literature regarding circuit topology, component selection, and waveform characteristics at the low-level circuit layer of the shuffling converter. Given this, the research project focuses heavily on these factors through low-level circuit simulation and prototype design and testing. The key questions identified for this research project are:

- What is an appropriate circuit topology for a shuffling converter, given the ever-increasing voltage and current output of modern PV modules?
- What negative effects (if any) does the power shuffler introduce, particularly when operating in extreme conditions?
- How much energy can be restored by the shuffling converter once losses are accounted for?
- How does the power shuffler behave when shuffling different amounts of energy between PV modules?

To answer these questions, deliverables have been identified across both semesters of the project. This report focuses primarily on those completed in the first semester and the implications of the results for the hardware design phase that follows. The primary outcome of semester one is this interim report, detailing results from low-level circuit simulations which inform the design decisions for the physical prototype. At the time of this report, all simulation files have been completed; the initial PCB schematic has been delayed due to component changes and by a dependency on the parallel control systems stream - both of which are documented in detail in Section 2.2 and managed via the buffer weeks allocated in the original timeline.

2. Interim Deliverables

Deliverables in the first semester revolve around three principal circuit simulations and their results: candidate shuffler topology comparison, the ideal shuffler model, and the semi-realistic shuffler model. Supporting sub-circuits, most notably the PV module model, have also been created to ensure realistic depiction of the three main circuits and to provide a like-for-like baseline for performance comparison against conventional bypass-diode mitigation.

For the purpose of accurate, low-level circuit simulation, Qspice was chosen due to its free cost, open-source nature, and native C++ integration, which better allows for control system integration compared to its predecessors. Being a SPICE language, Qspice is also backward compatible with older SPICE models such as those developed in LTspice, expanding the pool of validated component models available for simulation. Despite this, the software have yet to become more common in the control space compared to competitors such as MATLAB Simulink due to it being early in the development cycle and limited documentation on control systems.

2.1 Circuit Simulations

2.1.1 PV Module Modelling

To ensure that subsequent circuit simulations reflect the behaviour of real PV modules rather than an idealised approximation, an accurate PV module model was required in place of a simple ideal voltage source. For this purpose, the Karmalkar–Haneefa model (KHM) [16] was selected to represent each module's I–V behaviour. The KHM was chosen due to its simplicity in comparison to the single-diode model, which, whilst accurate, is computationally expensive and relies on parameters that are difficult to obtain. The KHM instead derives its parameters directly from datasheet values such as the voltage and current value in short circuit, open circuit and MPP conditions, making parameter extraction far more straightforward whilst still providing a far more realistic representation of module behaviour than an ideal voltage source.

The Aiko A540-MCE60Mw solar module was chosen to be modelled in Qspice due to it having one of the highest power outputs in the consumer PV market, allowing for accurate representation of a modern, high-output PV module. Evaluating the shuffler against this module also allows for worst-case scenario analysis, where the shuffler must handle a large amount of energy, such as in cases of severe shading mismatch.

The method used to build the KHM module in Qspice was adapted from Karmalkar and Haneefa [16]. The data required to parameterise the model was extracted from the module's I–V curve in the datasheet, using a digitiser script in MATLAB. Linear regression was then performed on the digitised curve to determine the model's fitting parameters, which is then used to parameterise the PV module model in Qspice. The model was validated by comparing the simulated P–V curve against the equivalent datasheet's graph at multiple irradiance levels of 1000, 800, 600 and 400 W/m², with strong agreement observed across the operating voltage range.

2.1.2 Shuffler Topology Selection

With a validated PV module model in place, three candidate shuffler topologies were evaluated to inform the final topology selection: the half-bridge buck-boost, the Ćuk converter, and the interleaved half-bridge buck-boost. These three topologies were identified in the project proposal as the most viable non-isolated bidirectional converters for the parallel-connected partial-power processing (P-PPC) role, drawing on the classification of Kasper et al. [7] and the foundational shuffler architecture of Walker et al. [13]. Inductor current ripple was identified at the outset of the design process as the dominant selection criterion and remained the deciding factor throughout: ripple amplitude sets the lower bound on the inductor and filter capacitor sizes, drives RMS current

stress on the active switches, and is the primary contributor to conducted-EMI performance once the converter is built. A topology that achieves low ripple at a given switching frequency therefore translates directly into smaller, cheaper, and quieter hardware.

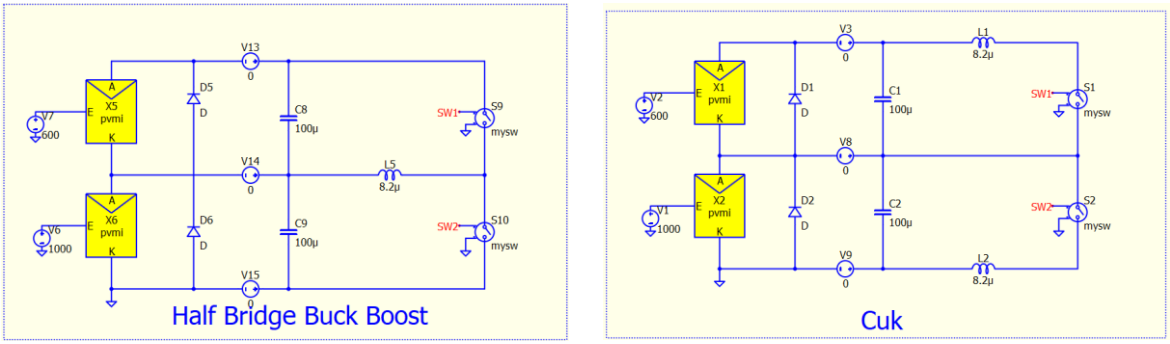


Figure 1 — The two unselected candidate topologies as implemented in Qspice: (a) the half-bridge buck-boost (left), and (b) the Ćuk converter (right).

The half-bridge buck-boost is the simplest of the three: two active switches and a single inductor carry the redistributed current between adjacent modules. Its appeal lies in low component count, but as a single-stage converter it produces the full inductor ripple at the converter input and output, with no mechanism to attenuate it other than increasing inductor size or switching frequency. The Ćuk converter uses two inductors and a coupling capacitor in series with the main current path; whilst this produces continuous input and output currents, its inductor ripple current is similar to that of a half-bridge buck boost converter.

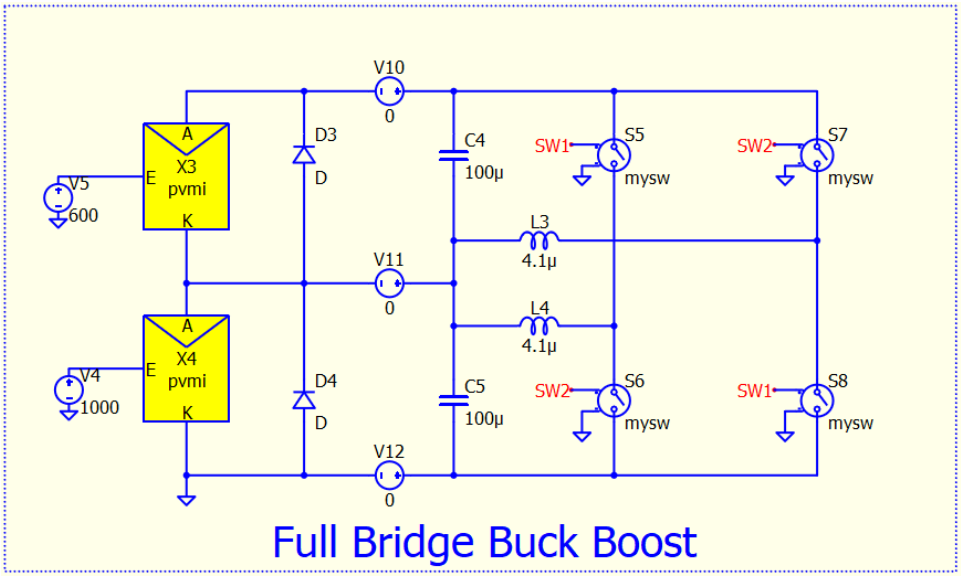


Figure 2 — The selected interleaved half-bridge buck-boost shuffler topology as implemented in Qspice..

The interleaved half-bridge buck-boost combines two half-bridge stages in parallel, each driving its own inductor, with the gate drives of the second stage phase-shifted by 180° relative to the first. The two inductor currents are equal and opposite in their ripple components but identical in their DC components, so the ripple cancels at the converter terminals whilst the redistributed DC current adds. The result is an effective input/output ripple that, at 50% duty cycle, approaches zero and remains substantially below either single-stage topology across the full duty range without any increase in inductor size or switching frequency.

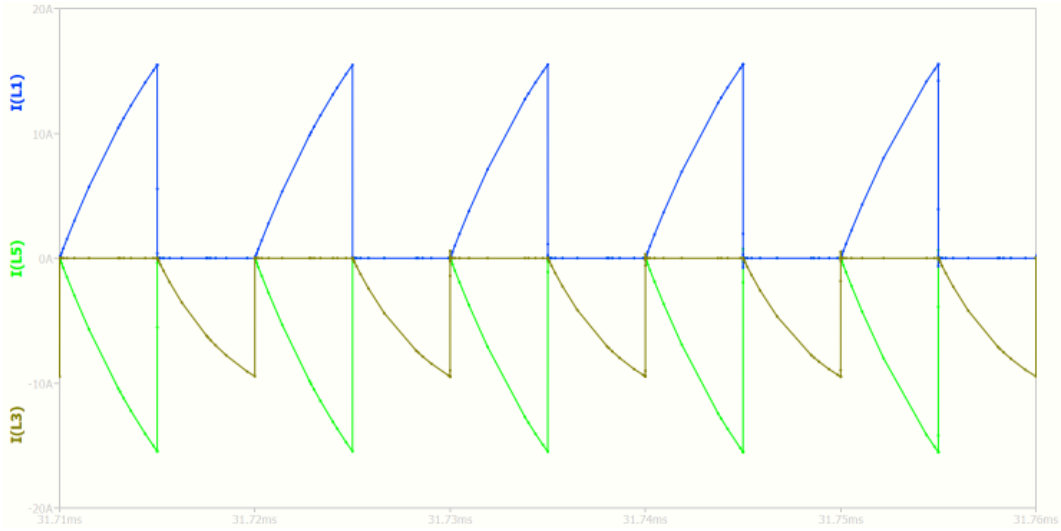


Figure 3 — Inductor current waveform comparison between the three candidate topologies operating under identical conditions. The half-bridge buck-boost and Ćuk converter (blue and green traces) exhibit substantially higher ripple magnitude than the interleaved buck-boost (brown trace, single inductor shown for clarity). The phase cancellation in the interleaved variant is the basis of its selection.

On the basis of inductor current ripple alone, the interleaved half-bridge buck-boost was selected to carry forward to the hardware design phase. The waveform comparison in Figure 3 makes the case directly: under identical operating conditions, the two single-stage topologies carry inductor ripple amplitudes is double that of the interleaved variant. Matching the interleaved topology's ripple performance with either of the single-stage alternatives would require significantly larger inductors (with associated cost, size, and core-loss penalties) or a higher switching frequency (with associated switching-loss penalties), neither of which is consistent with the scalable, cost-effective design philosophy established in the project proposal.

2.1.3 Ideal Shuffler Model

Following topology selection, a detailed ideal model of the interleaved buck-boost shuffler was constructed in Qspice to determine the converter's behaviour without device-level losses. In this model, all active switches are represented by ideal voltage-controlled switches with zero resistance and instantaneous switching time with gate drivers are modelled as ideal voltage sources. This simulation allows for analysis of the converter's performance (inductor and capacitor ripple) and the upper bound of recovered energy levels to be evaluated without confounding contributions from real-device losses. Thus, a providing a clean ceiling which the semi-realistic model in Section 2.1.4 can be benchmarked against. The ideal model is shown in Figure 3.

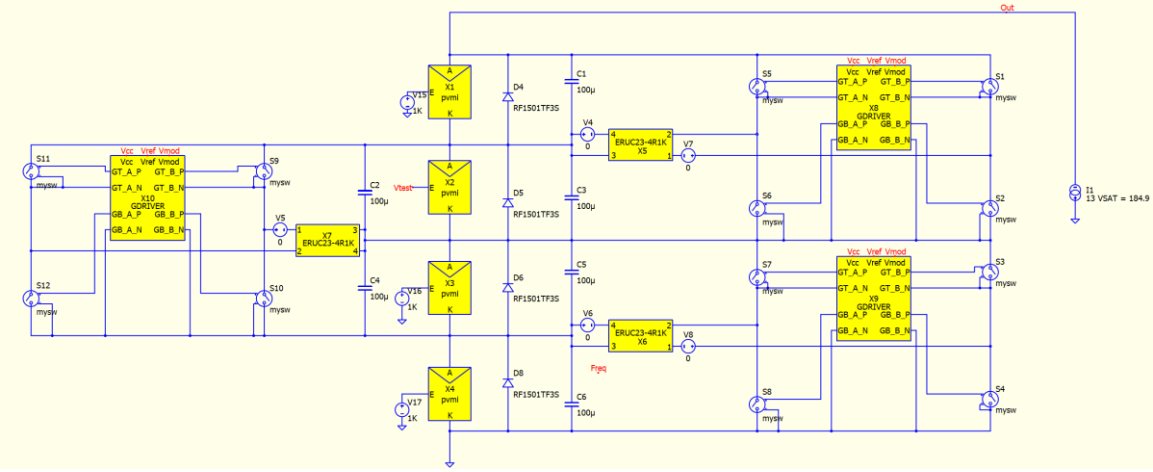


Figure 3 — Implementation of the ideal interleaved buck-boost topology in Qspice.. PV modules X1–X4 use the KHM model from Section 2.1.1.

To produce a like-for-like comparison, two simulations were run on the same PV string under identical shading conditions: the first with bypass diodes only (representing standard industry practice) and the second equipped with ideal shuffle converters. The output power of each configuration was recorded over the duration of the shading event, and the instantaneous difference between the two was computed to yield the recovered power. The result is shown in Figure 4.

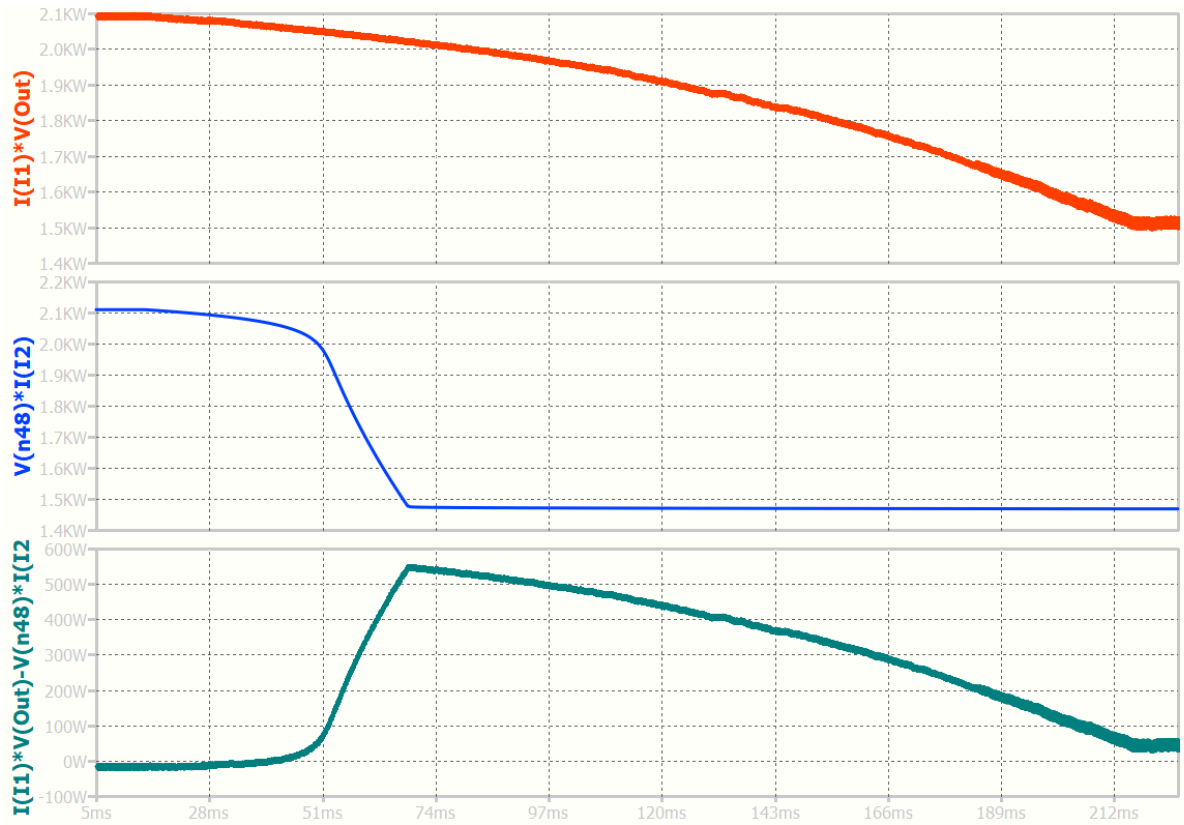


Figure 4 — Energy recovery comparison from the ideal model. Top: string output power with the shuffler active (red); middle: string output power with bypass diodes only (blue); bottom: instantaneous recovered power (teal). Peak recovery of approximately 525 W occurs at ~70 ms.

Prior to the shading event (0–15ms), both configurations produce identical output power of approximately 2.1 kW, confirming that the shuffler introduces no measurable parasitic loss under

uniform irradiance conditions. This is a direct consequence of the partial-power processing architecture: under matched module conditions, no current flows through the shuffler and it effectively acts as a transparent element in the string.

At approximately 15ms, a gradually worsening shading event is introduced. At very light shading conditions, both PV strings have similar power output. However, at ~45ms, the bypass diode begins to activate, removing the shaded module's contribution from the traditional string entirely. The shuffler configuration, by contrast, redistributes current to the shaded module rather than bypassing it, holding the string output substantially closer to its pre-shading level. The instantaneous recovered power peaks at approximately 525 W shortly after the shading onset, before gradually declining as the simulation progresses and the shading scenario deepens to a final value of 200W/m^2. By the end of the simulation window, the shuffler's recoverable advantage has diminished as the configuration approaches the limit of what can be redistributed within the string.

This result addresses Research Question 3 directly: under the simulated conditions and in the ideal limit, the interleaved full-bridge shuffler recovers a peak of approximately 525 W of power that would otherwise be lost to bypass action. This represents a substantial fraction of the deficit and a quantitative justification for the partial-power shuffler architecture. The decline in recovered power over the shading event is itself an informative result, as it identifies the operating region beyond which the shuffler can no longer fully compensate. In practice, this means that if a PV module is operating in extreme shading conditions, the shuffler connected to that module should be switched off and allowing for the bypass diode to operate as usual to reduce component stress.

2.1.4 Semi-Realistic Shuffler Model

The ideal model establishes an upper bound on shuffler performance but does not capture the losses associated with real semiconductor devices, which are the dominant efficiency consideration for any practical implementation. To assess what fraction of the ideal recovery survives the introduction of real devices, a semi-realistic model was constructed by replacing the ideal switches and gate drivers of the ideal model with their realistic counterparts. The ideal switches were replaced with MOSFETs generated using Qspice's component generator and the manufacturer's datasheet, capturing on resistance (R_{ds_on}) gate charge (C_{oss}), body diode behaviour, and switching transition times. Ideal gate drivers were replaced with a behavioural gate driver model that produces realistic gate voltage waveforms, including rise and fall times and the dead-time interval between complementary switches. Lastly, parasitic series resistances were added to all components to account for unavoidable losses and eliminate unrealistic behaviour such as large, instantaneous current discharge from capacitors in the ideal model. To prevent excessive simulation time and unrealistic switching nodes voltage ripple, components and PCB traces/ parasitic inductance were omitted from this model, hence the naming convention of 'semi-realistic' model.

The semi-realistic model is shown in Figure 5.

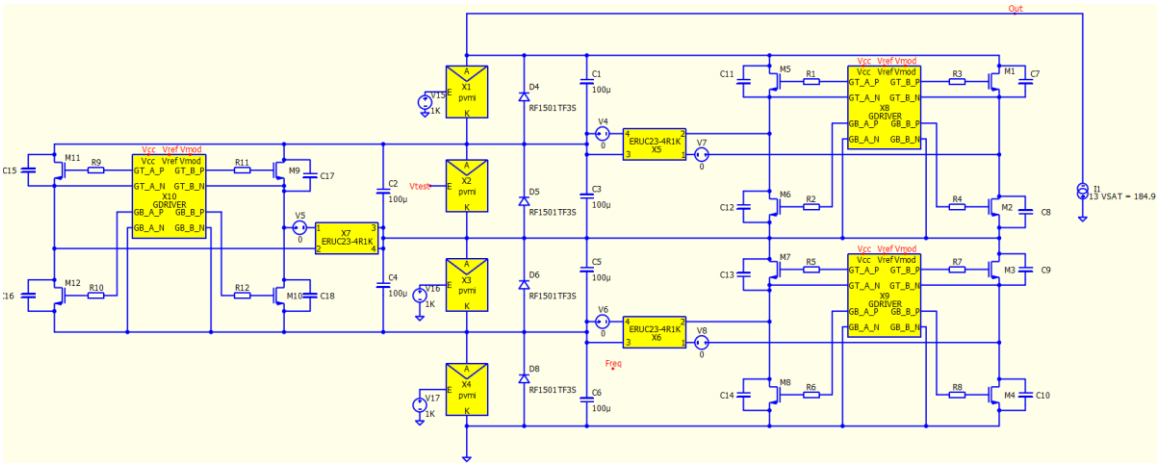


Figure 5 — Semi-realistic shuffler model

Two principal insights emerged from the semi-realistic model. The first is the introduction of switching and conduction losses, which slightly reduce the net energy recovery below the ideal-case peak of 525 W identified in Section 2.1.3 to approximately 515W. When operating in extreme condition (one model receiving 200W/m^2 of irradiance), there is an additional 50-60W of losses in comparison the ideal model. Despite this, the shuffler retains a meaningful net advantage over the bypass baseline, with the loss tax falling within the range typical of well-designed buck-boost converters operating at the targeted switching frequency for most of the load range. The result of the semi realistic model could be seen in Figure 6.

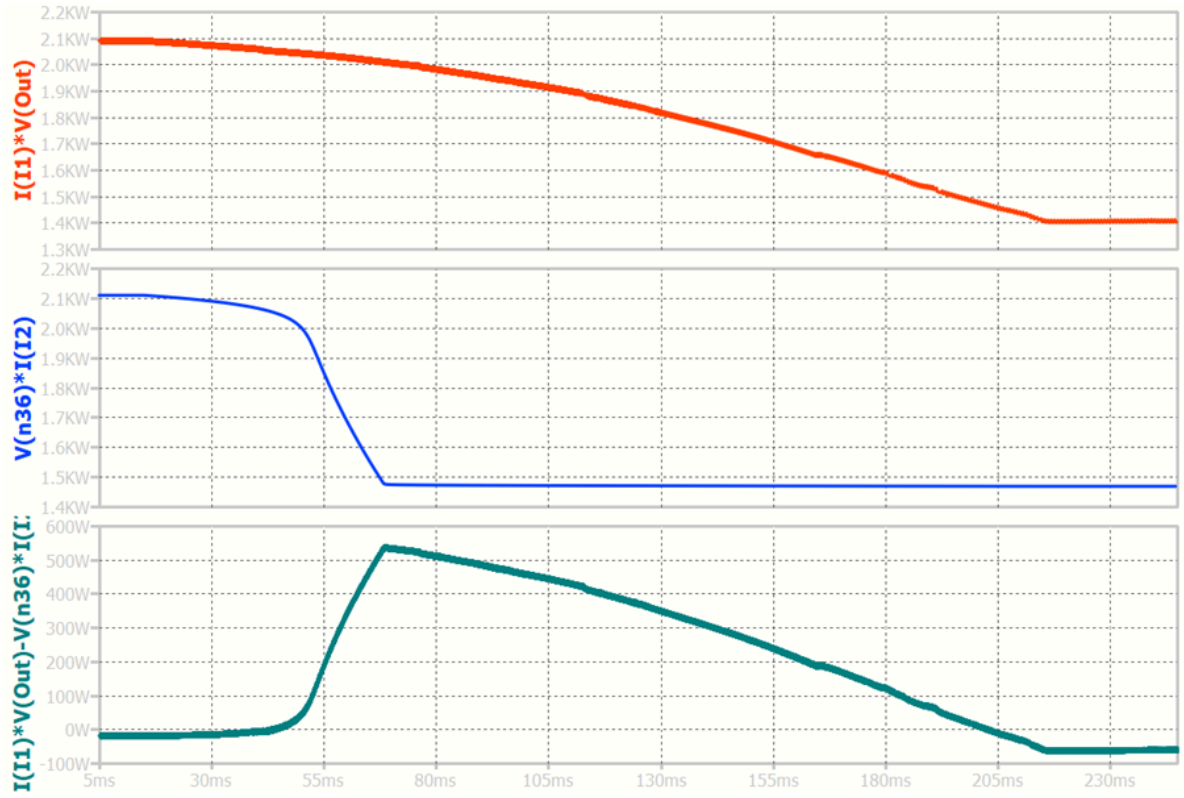


Figure 6 — Energy recovery comparison from the semi-realistic model

The second and more design-relevant insight concerns dead-time control. In the ideal model, complementary switches in each full-bridge leg can be commutated instantaneously with no overlap, as the switches themselves have zero transition time. In the semi-realistic model, a finite

dead-time interval is required between complementary switches to prevent shoot-through, but a fixed dead-time value performs poorly across the full operating range of the shuffler. This could be observed with lower recovered energy levels at light (-15 W) and heavy shading (-50-60 W). At light shuffling loads, excessive dead-time results in body-diode conduction and elevated conduction losses; at heavy shuffling loads, insufficient dead-time led to shoot-through and, causing thermal higher stress on the switch and lowering efficiency. The semi-realistic simulations confirm that a dynamic dead-time control scheme, which adjusts the dead-time interval based on the current operating condition, is required to maintain efficient operation across the shuffler's full load range. This finding directly informs the control system design being developed in parallel by the project's controls stream, and represents a concrete change in scope at the interface between the two streams.

2.2 Review and Identified Challenges

Overall, the simulation phase of the project progressed substantially in line with the deliverables agreed upon at the start of semester. A validated PV module model was implemented; three candidate shuffler topologies were simulated and compared; the interleaved buck-boost was selected as the topology to carry forward; and both an ideal and a semi-realistic model of the chosen topology were constructed and produced informative results, with peak energy recovery quantified at approximately 515-525 W under the simulated shading scenario.

Two challenges have impacted the originally planned delivery of the initial PCB schematic within semester one. The first is a design change from a $250\mu H$ inductor to a coupled $4.7\mu H$ inductor pair for smaller footprint size, lower winding losses and increased inductor ripple current tolerances. As a result of this change, the switching speed was increased from $50kHz$ to $200kHz$ to ensure manageable voltage ripple on the input/output capacitors. The high-level redesign is currently ongoing, with component selection and initial hardware layout mostly completed.

The second challenge is a dependency on the control systems stream. The first PCB revision requires microcontroller pinout and peripheral requirements from the controls student to ensure the board can support the agreed-upon control interface; these requirements were not finalised within semester one and have been carried into the summer break. To mitigate the impact, the PCB schematic for the power stage is to be progressed in parallel where possible; component selection, gate driver circuits, and current sensing have been largely laid out, with only the microcontroller interface section left as a placeholder to be populated when the controls requirements are received. This approach is consistent with the buffer-week allocation in the original project timeline and is not expected to affect the semester two prototype build schedule. A brief summary of the resulting timeline adjustment is provided in Appendix A.

2.3 Limitations and Assumptions

Several simplifying assumptions underpin the simulation results presented above. Acknowledging these explicitly is important both for honest reporting and for scoping the remaining semester two work.

Shading scenario specificity. The energy recovery figure of 525 W is derived under a specific simulated shading scenario applied to a four-module string. Real-world shading patterns such moving cloud cover, intermittent obstructions, and partial-cell shading may produce recovery values either above or below this figure depending on the depth and distribution of the mismatch. The 525 W figure should be interpreted as an order of magnitude indication of recoverable power under representative conditions rather than as a system-wide performance guarantee.

Device modelling fidelity. The semi-realistic model captures conduction losses, switching losses, and dead-time effects, but does not incorporate parasitic inductance, thermal effects, magnetic saturation in the inductors at peak shuffling currents, or parasitic inductance and capacitance from PCB layout. These second-order effects will be quantified during prototype bench testing in semester two and may further reduce the achievable net recovery.

Shading event dynamics. The shading event in the simulation is introduced as a gradual change in irradiance on the affected module. Real shading transitions, such as for use in applications such as solar cars, can occur rapidly. The shuffler's tracking dynamics over these step input like conditions are not yet characterised. The Perturb and Observe MPPT algorithm planned for the controls stream will need to be validated against fast irradiance transients during semester two.

3. Risk Management

The risk register established in the project proposal has been reviewed in light of the semester one experience, with several risks reassessed in either likelihood or impact. The discussion that follows separates non-technical project risks from experimental risks anticipated for semester two, and distinguishes between risks that have materialised, risks whose likelihood has shifted, and risks that remain dormant.

Project Risks

The risks originally identified as scope creep, electrical safety during testing, component lead-time delays, and design interdependency between the power electronics and control systems streams have all evolved through the semester. Scope creep has not materialised: the simulation deliverables agreed in the proposal have been completed in their original form, and no new scope items have been added. The decision to defer detailed thermal and EMI characterisation to the prototype phase is an explicit scope boundary rather than a slip and was made on the basis that these effects are inadequately captured in simulation regardless of model fidelity.

The design interdependency risk, by contrast, has materialised in the form described in Section 2.2. The original mitigation strategy of agreeing interface specifications early is proved insufficient as it did not sufficiently account for large delays in deliverables from one stream. The updated mitigation is structural rather than procedural: the PCB layout now occur side by side with the control system design stage to allow power-stage design to progress more independently. This change applies to the version-one PCB and will be preserved in subsequent revisions. The lesson of allowing for more independent operations between the power electronic and control system sides has been documented as a project risk-management insight for the remaining hardware phase.

Component lead-time risk has not yet been tested as no components have been ordered. However, with component ordering scheduled for the early weeks of the summer break, this risk shifts from dormant to active and will be monitored closely. The risk profile for the remainder of the project shifts from primarily software and simulation risks to hardware risks, particularly around PCB revision cycles, gate driver layout, and the elevated voltages anticipated during live PV string testing in semester two.

Experimental Risks

Experimental risks were minimal during semester one as all work was conducted in simulation and posed no electrical, thermal, or material hazards. The risk profile changes substantially in semester two with the introduction of bench testing and subsequent live PV string testing. Bench testing of individual shuffler PCBs will be conducted using programmable DC power supplies emulating PV source behaviour, keeping voltages and currents within the low-voltage envelope and removing the

risk of working with live solar irradiance during initial validation. The incremental testing protocol established in the proposal - starting from low-power bench tests and progressing to live panel tests only after each preceding stage is validated - will be followed to prevent escalation of fault conditions.

Once individual unit testing is complete, integration with a real PV string will be conducted at QUT's Power Engineering Research Facility under the supervision of a licensed electrical technician, in compliance with the AS/NZS 3000 Australian Wiring Rules [17]. The PV string voltage may exceed 150 V DC, placing the work above the 120 V DC threshold defined in the Queensland Electrical Safety Act 2002, which makes licensed supervision a regulatory rather than discretionary requirement. The semi-realistic model's findings on dead-time control add a further risk dimension: an incorrectly configured dead-time at heavy load can cause shoot-through and immediate switch destruction, with associated arc-flash risk at the PV string voltage level. Test procedures will therefore include staged duty-cycle ramps beginning at light loads, and the gate driver hardware will incorporate hardware-level dead-time enforcement as a backstop to the firmware-level dynamic dead-time control. This represents a substantive design change introduced as a direct result of the semester one simulation findings.

4. Consideration of Ethics

Ethical decision-making throughout the project has been guided by the Engineers Australia Code of Ethics and Guidelines for Professional Conduct [18], which articulates four core principles: demonstrate integrity, practise competently, exercise leadership, and promote sustainability. Each principle has been considered against the specific context of this project, with particular attention to decisions where the ethical and technical considerations were not trivially aligned.

Demonstrate Integrity

Integrity has been observed most visibly in the accurate reporting of simulation results where results have been reported honestly, including the as-yet-unresolved convergence issue in the semi-realistic model; where results are preliminary or contingent on further work, this is explicitly noted rather than implied to be complete. The limitations section (2.3) is part of the same commitment — acknowledging the bounds of validity of the simulation results rather than presenting them as universally applicable.

Integrity also extends to the proper attribution of external intellectual contributions. The Karmalkar–Haneefa PV model implementation [16] is externally-developed work; the project's own contribution is the instantiation of the model in Qspice, its parameterisation against the Aiko A540-MCE60Mw datasheet, and its integration into the broader shuffler simulation. This distinction is explicit in Section 2.1.1 to avoid any implication that the model itself is a project contribution. Similarly, the partial-power processing shuffler architecture itself originates from Walker et al. [13]; the project's contribution is the low-level circuit, component, and waveform characterisation that the existing literature on this architecture has not yet addressed in depth.

Practise Competently

Competence has been pursued through the application of established engineering tools and validated methodologies rather than ad-hoc approaches. Qspice was chosen as the simulation environment for its open-source nature and compatibility with established SPICE conventions, ensuring that results are reproducible and that component models are drawn from validated sources. The shuffler topologies investigated are drawn directly from the published literature on differential power processing and partial-power converters [7], [13], [15], ensuring that the project advances within an established research framework rather than reinventing it. The classification of converter

topologies by Kasper et al. [7] was used directly to scope the project's design space, and the topology selection was made against the same criteria that the literature uses for evaluation.

Competent practice is also demonstrated in the explicit recognition of project limits. The live PV testing planned for semester two is acknowledged as exceeding the unlicensed electrical work threshold under the Queensland Electrical Safety Act, and accordingly will be conducted under the supervision of a licensed electrical technician at QUT's Power Engineering Research Facility. The decision to defer detailed thermal modelling to the prototype phase is similarly a competence-based judgement - simulation models without PCB-level thermal coupling produce thermal predictions that are misleading rather than informative.

Exercise Leadership

The project is conducted collaboratively with a parallel control systems stream, and leadership has been exercised through proactive communication at the stream interface. When the design dependency caused friction - particularly in the microcontroller interface requirements - the response was to escalate the issue early to the supervisor and to restructure the PCB design approach to allow independent progress on the power-stage components. This approach prioritises the overall project outcome over the apportionment of responsibility between the two streams, and reflects the Engineers Australia expectation that engineers communicate honestly and act responsibly within a team context. The dynamic dead-time finding from the semi-realistic model (Section 2.1.4) was also escalated to the controls stream as a substantive interface change rather than absorbed silently within the power electronics scope, since the firmware implications fall within the controls stream's responsibility.

Promote Sustainability

Sustainability considerations are documented in detail in Section 5. From an ethical standpoint, the project's overall direction of improving the energy yield of existing PV installations rather than promoting the expansion of new ones, represents a deliberate alignment with sustainability principles. Decisions on component selection and PCB revision strategy have been made with longevity, repairability, and waste minimisation in mind, as elaborated below.

Intellectual Property Considerations

An additional ethical consideration arises from the existence of commercial products in the power optimisation space, notably those produced by SolarEdge, which hold intellectual property in related architectures. As this project is conducted for academic research purposes and not for commercial deployment, IP conflict risk is considered low. However, care has been taken in the documentation to distinguish the novel aspects of this design - particularly the low-level circuit, component, and waveform characterisation of the interleaved full-bridge shuffler from existing patented full-power optimiser implementations. The architectural distinction between partial-power processing (this project) and full-power processing (commercial optimisers) is itself a substantive technical differentiator that has been preserved throughout the design and documented in the literature review section of the proposal.

5. Implementation of Sustainable Development Principles

Sustainable development considerations have been incorporated throughout the project design, guided by the Engineers Australia document Implementing Sustainability: Principles and Practice [19]. The most pertinent guidelines have shaped specific design decisions as discussed below, and where possible the sustainability rationale is connected to a quantitative or structural design choice rather than a generic statement of intent.

System-Level Sustainability through Improved Energy Yield

The most significant sustainability contribution of this project is at the system level: by improving the energy extraction efficiency of existing PV installations under partial shading conditions, the shuffler reduces the need for additional panel installations to achieve equivalent energy output. This addresses EA sustainability guidelines 1 and 7 by promoting more effective use of an established renewable resource and by offering a proactive, integrated solution to a known inefficiency in the existing PV layout rather than relying on the reactive expansion of capacity. The 525 W peak recovery quantified in Section 2.1.3 is a useful order-of-magnitude indication. For context, a typical residential rooftop installation experiencing recoverable partial shading for several hours per day during morning and evening shoulder periods could see cumulative annual recovery in the hundreds of kWh. This represents an amount of energy that would otherwise require additional panel area to generate, with the associated embodied energy and material cost of those additional panels.

Component Selection for Longevity and Repairability

Component selection prioritises industrial and automotive-grade parts for critical elements such as the power MOSFETs, gate drivers, and electrolytic capacitors, which are the dominant lifetime-limiting components in DC–DC converters. Industrial grade parts typically offer extended operational lifetimes under thermal and electrical stress compared with consumer-grade equivalents, addressing EA sustainability guideline 4 (product design longevity, re-use, repair, and recyclability). Where industrial grade parts are not justifiable on a cost basis for the prototype phase, the design ensures that the parts can be substituted later without redesigning the surrounding circuit. The use of socket-mounted or commonly-packaged ICs in the prototype PCB will also support repair and component swapping during the bench testing phase, reducing the rate at which boards are discarded due to individual component failure.

Modular Hardware Strategy

The project's planned three-revision PCB strategy is intentional from a sustainability standpoint. Rather than discarding entire assemblies when issues are identified, each revision is designed to incorporate lessons from the previous one whilst keeping the core power-stage layout and component selection consistent where possible. This addresses EA sustainability guideline 5 (eliminating waste as a primary design consideration) by avoiding wholesale redesign and by enabling components harvested from earlier revisions to be reused for diagnostic purposes on later revisions.

Standardised Interfaces and Long-Term Compatibility

Connectors, terminals, and communication interfaces have been selected to be industry-standard where practical (e.g. XT-60), ensuring long-term replaceability and interoperability without requiring full system redesign. The use of a topology-agnostic mechanical footprint also extends the useful life of the hardware by avoiding obsolescence tied to any specific PV panel generation, addressing EA guideline 3 (whole-system design consideration from inception to reuse). Sizing the shuffler against the high-output Aiko A540 module also ensures compatibility with lower-rated modules currently in the field, extending the addressable installed base over which the design can deliver energy-recovery benefits.

6. Resources, Codes, and Standards

Software

Qspice was selected as the primary simulation environment for its open-source nature, native C++ integration (which enables co-simulation of the control system in semester two without proprietary

tooling), and backward compatibility with the LTspice model library. KiCad is the intended PCB design tool, again selected for its open-source nature and broad community support, and to avoid tying the design to commercial EDA software licences that could not be carried into post-graduation work. MATLAB was used in supporting roles for I–V curve digitisation and linear regression of the KHM model parameters; Python is planned for post-processing of experimental data in semester two.

Hardware and Facilities

Bench testing in semester two will utilise programmable DC power supplies and electronic loads to emulate PV source and MPPT load behaviour, oscilloscopes for switching waveform inspection, and IR imaging equipment for thermal characterisation of the prototype under sustained load. Live PV string testing will be conducted at QUT's Power Engineering Research Facility, which provides access to multichannel data acquisition systems and power analysers required for evaluating shuffling current between modules in a real string configuration.

Codes and Standards

All hardware design and testing must comply with AS/NZS 3000:2018 — the Australian/New Zealand Wiring Rules [17]. Where the PV string voltage exceeds 120 V DC (which is anticipated during live testing on the prototype string), the Queensland Electrical Safety Act 2002 requires that the work be conducted by, or under the supervision of, a licensed electrical technician. PV module specifications are drawn from manufacturer datasheets, with the Aiko A540-MCE60Mw selected as the reference module for design sizing per Section 2.1.1. Engineers Australia's Code of Ethics [18] and Implementing Sustainability guidelines [19] underpin the discussion in Sections 4 and 5.

7. Stakeholder Engagement

The primary project stakeholder is the academic supervisor, Associate Professor Geoff Walker, with whom weekly progress meetings have been held throughout the semester where possible. These meetings have served both as technical review checkpoints, particularly at the completion of each major simulation deliverable, and as the channel through which design decisions such as the topology selection have been validated. Walker's foundational research on per-module MPP converters [13] has directly shaped the architectural approach taken by this project, and the project's focus on low-level circuit, component, and waveform characterisation aligns directly with a gap that Walker has flagged as under-addressed in the existing literature on shuffler converters.

The control systems stream, led by fellow student engineer Jared Saunders, is a second active stakeholder due to the design interdependency between the two streams. Interface-defining decisions, particularly the microcontroller selection, communication protocol between shuffler units, and the dead-time control implementation, have been the subject of explicit discussion between the two streams. The dependency-related delay described in Section 2.2 has prompted an increase in the frequency of these check-ins for the remainder of the project.

QUT's Power Engineering Research Division is an institutional stakeholder, as the facility and its equipment will be utilised for live PV string testing in semester two. Engagement with the facility has, to date, been confined to confirming availability and establishing the requirement for licensed supervision during testing. Further engagement will commence ahead of the first scheduled test session, including an induction with the licensed supervising electrician and a review of the test protocol against the facility's standard procedures.

8. Conclusion

Semester one of this project has delivered a complete and validated simulation foundation for the development of an interleaved full-bridge buck-boost shuffling converter. A datasheet-driven Karmalkar–Haneefa PV module model was implemented in Qspice and validated against expected non-linear behaviour. Three candidate shuffler topologies were systematically compared and the interleaved full-bridge buck-boost selected to carry forward on the basis of low switch stress, ripple cancellation through phase-shifted operation, and modular scalability. Both an ideal and a semi-realistic model of the chosen topology were constructed, with the ideal model demonstrating a peak energy recovery of approximately 525 W over the bypass-diode baseline under the simulated shading conditions, and the semi-realistic model confirming that dynamic dead-time control is a necessary feature of any practical implementation.

The initial PCB schematic, originally planned for completion within semester one, has been delayed due to design changes and by a dependency on the parallel control systems stream; both issues are documented in detail in Section 2.2 and have been planned for resolution within the early weeks of the summer break. The project remains on track for the prototype build, bench testing, and live PV string validation deliverables scheduled for semester two, with buffer-week allocation absorbing the slip. The low-level circuit and waveform characterisation produced this semester - an area identified in the literature review as comparatively under-explored in the shuffler research space - provides the design basis upon which the hardware phase of the project will proceed.

References

[1] IEA, "Snapshot of Global PV Markets," 2025. [Online]. Available: https://www.iea.org/reports/world-energy-outlook-2025

[2] Al Bari and Shahool, "Solar Electricity Generation in Australia," IBISWorld, Jun. 2025. [Online]. Available: https://my.ibisworld.com/us/en/industry/22111e/performance

[3] D. K. Narne, T. A. Ramesh Kumar, and R. K. R. Alla, "A novel PV array configuration for enhancing maximum power from PV array," Clean Energy, vol. 6, no. 6, pp. 817–826, Dec. 2022, doi: 10.1093/ce/zkac054.

[4] B. Aljafari, S. Devakirubakaran, C. Bharatiraja, P. K. Balachandran, and T. S. Babu, "Power enhanced solar PV array configuration based on calcudoku puzzle pattern for partial shaded PV system," Heliyon, vol. 9, no. 5, e16041, May 2023, doi: 10.1016/j.heliyon.2023.e16041.

[5] T. Ahmad, S. Sobhan, and Md. F. Nayan, "Comparative Analysis between Single Diode and Double Diode Model of PV Cell," Journal of Power and Energy Engineering, vol. 4, no. 3, pp. 31–46, 2016, doi: 10.4236/jpee.2016.43004.

[6] O. Bingöl and B. Özkaya, "Analysis and comparison of different PV array configurations under partial shading conditions," Solar Energy, vol. 160, pp. 336–343, Jan. 2018, doi: 10.1016/j.solener.2017.12.004.

[7] M. Kasper, D. Bortis, and J. W. Kolar, "Classification and Comparative Evaluation of PV Panel-Integrated DC–DC Converter Concepts," IEEE Transactions on Power Electronics, vol. 29, no. 5, pp. 2511–2526, May 2014, doi: 10.1109/TPEL.2013.2273399.

[8] R. I. Rico-Camacho, L. J. Ricalde, A. Bassam, M. I. Flota-Bañuelos, and A. Y. Alanis, "Transient Differentiation Maximum Power Point Tracker (Td-MPPT) for Optimized

Tracking under Very Fast-Changing Irradiance," Applied Sciences, vol. 12, no. 5, p. 2671, Mar. 2022, doi: 10.3390/app12052671.

[9] T. Esram and P. L. Chapman, "Comparison of Photovoltaic Array Maximum Power Point Tracking Techniques," IEEE Transactions on Energy Conversion, vol. 22, no. 2, pp. 439–449, Jun. 2007, doi: 10.1109/TEC.2006.874230.

[10] P. Fernández-Bustamante, I. Calvo, E. Villar, and O. Barambones, "Centralized MPPT based on Sliding Mode Control and XBee 900 MHz for PV systems," International Journal of Electrical Power & Energy Systems, vol. 153, p. 109350, Nov. 2023, doi: 10.1016/j.ijepes.2023.109350.

[11] I. M. Mehedi et al., "Critical evaluation and review of partial shading mitigation methods for grid-connected PV system using hardware solutions," Renewable and Sustainable Energy Reviews, vol. 146, p. 111138, Aug. 2021, doi: 10.1016/j.rser.2021.111138.

[12] O. Khan, W. Xiao, and H. H. Zeineldin, "Gallium-Nitride-Based Submodule Integrated Converters for High-Efficiency Distributed MPPT PV Applications," IEEE Transactions on Industrial Electronics, vol. 63, no. 2, pp. 966–975, Feb. 2016, doi: 10.1109/TIE.2015.2491888.

[13] G. R. Walker, J. Xue, and P. Sernia, "PV string per-module maximum power point enabling converters," in Australasian Universities Power Engineering Conference, AUPEC'03, 2003.

[14] N. G. F. dos Santos, J. R. R. Zientarski, and M. L. da S. Martins, "A Two-Switch Forward Partial Power Converter for Step-Up/Down String PV Systems," IEEE Transactions on Power Electronics, vol. 37, no. 6, pp. 6247–6252, Jun. 2022, doi: 10.1109/TPEL.2021.3138299.

[15] C. Olalla, C. Deline, and D. Maksimovic, "Modeling and simulation of conventionally wired photovoltaic systems based on differential power processing SubMIC-enhanced PV modules," in IEEE 15th Workshop on Control and Modeling for Power Electronics (COMPEL), 2014, doi: 10.1109/COMPEL.2014.6877127.

[16] S. Karmalkar and S. Haneefa, "A Physically Based Explicit J–V Model of a Solar Cell for Simple Design Calculations," IEEE Electron Device Letters, vol. 29, no. 5, pp. 449–451, May 2008, doi: 10.1109/LED.2008.920466.

[17] Standards Australia, AS/NZS 3000:2018, Electrical Installations (Australian/New Zealand Wiring Rules).

[18] Engineers Australia, Code of Ethics and Guidelines on Professional Conduct, 2022.

[19] Engineers Australia, Implementing Sustainability: Principles and Practice, 2015.

Appendix A — Updated Timeline Status

All semester one simulation deliverables identified in the project proposal timeline (power electronics stream, Weeks 1–13) have been completed on schedule, with the exception of the initial PCB schematic which has been deferred to Weeks 1–2 of the summer break. Reasons for the deferral and the associated mitigation are documented in Section 2.2 of this report. The deferral is absorbed within the buffer-week allocation in the semester two schedule and does not affect downstream milestones, including PCB V2 design (Sem 2, Weeks 4–6), full-system validation (Sem 2, Weeks 7–10), or live PV string testing at QUT's Power Engineering Research Facility (Sem

2, Weeks 11–13). The detailed Gantt chart remains as published in Appendix A of the project proposal.